Siao Xu

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Research Fields

Machine Learning, **Panel Data**, **Financial Econometrics**, **Network Science**, **Applied Microeconomics and Empirical Finance**.

Education

Universität Mannheim, Germany *Sep 2021 – 2027(Expected)*
Ph.D. in Economics

Hong Kong University of Science and Technology, Hong Kong *Sep 2016 – Jun 2018*
M.Sc. in Financial Mathematics

London School of Economics, UK *Sep 2012 – Jun 2015*
B.Sc. in Mathematics and Economics

Job Market Paper

[1] Mixed Membership Estimation in Partial Correlation Network
with Endong Wang, under review

Abstract: Latent group structure is a standard device for modeling cross-sectional dependence in large panels, yet existing partial-correlation community methods typically assign each unit to a single group and do not distinguish the formation of conditional-dependence links from their strength. We propose a two-path mixed-membership partial-correlation network (2PMM-PCN). Mixed membership governs link formation along an extensive margin, while node-specific sociability governs link strength along an intensive margin. Although the two paths have distinct economic interpretations, their node-specific components enter the population signal only through a common scale, allowing the mixed memberships to be recovered by a single spectral procedure. We develop PartialCorr-Mixed-SCORE and establish a convergence rate for the average membership estimation error under weak dependence. In an application to U.S. state employment growth, the estimated memberships reproduce the broad group structure found in earlier studies while revealing economically meaningful overlaps that a single-group assignment cannot represent.

Working Paper

[2] Mixed Membership Estimation in Large Network Autoregression
with Endong Wang, *R&R* at The Econometrics Journal

[3] The Horizon Structure of Contagion: High-Dimensional Measurement and Inference
with Abderrahim Taamouti and Endong Wang

[4] Overlapping Community Detection in Mixed Membership Vector Autoregression

[5] Machine Learning Assisted Empirical Bayes Estimation for Group Regression on Network Data

Work in Progress

[1] **Bi-group Detection in Large Matrix-Variate Factor Model**
with Endong Wang

[2] **Mixture Prior on Random Coefficient Model**

Employment

ShanghaiTech University, China *Nov 2018 – Jun 2021*
Research Assistant

Loan analyst *Sep 2015 – Jun 2016*
Industrial bank, China

References

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Universität Mannheim
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Universität Mannheim
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Prof. Mengshan Xu
Department of Economics
Universität Mannheim
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Conference, Workshop and Seminar

[**2026**] 2026 CSW Asia Meeting (Main) in Abu Dhabi, ENTER Jamboree in Madrid, Mannheim Econometrics Seminar, Time Series AI in Macroeconomics and Finance in Bristol, Bristol Econometric Study Group Conference, 31st International Panel Data Conference in Exeter, 2026 European Summer Meeting (ESEM) in Dublin, UCL PHD seminar.

[**2025**] (EC^2) Conference in Lugano, 45th International Symposium on Forecasting, 9th SETA (Macau), Mannheim Econometrics Seminar, HKMetrics Workshop in Mannheim, 2025 Harbin International Conference on Econometrics and Statistics, GESS Research Day in Mannheim.

[**2024**] Workshop in Asian Summer School of Econometrics, Mannheim Econometrics Seminar.

Teaching

[**Fall 2026**] CC502 Applied Econometrics (Master Level), Universität Mannheim

[**Fall 2025**] BE510 Business Economics 1 (Master Level), E700 Mathematics for Economists (Doctor Level), Universität Mannheim

[**Fall 2023**] BE510 Business Economics 1 (Master Level), Universität Mannheim

[**Spring 2023**] Macroeconomics A (Bachelor Level), Universität Mannheim

[**Fall 2022**] CC502 Applied Econometrics (Master Level), Universität Mannheim

Technologies

Programming: Python, R, Matlab, Stata, C++, C, Java

Language: Mandarin (native), English (fluent)

Nationality: Chinese